

Cloud Native Streaming Platform Netflix-inspired

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1 Executive Summary

Netflix is a streaming platform that provides this service for subscriptions. Whose vision is entertaining customers through series and movies. This content can be accessed around the world by millions of customers those who want to watch their favorites shows at any time. For that reason is an application demands great availability, good performance with heavy content such as multimedia and personalized experience with assertive recommendations.

2 System Description

Business problem, actor catalog, use case definitions, and system boundary diagram

2.1 Business Problem

Netflix solves the problem of providing a large amount of audiovisual content to millions of users through the internet while giving each user a convenient and personalized experience.

Users want to watch movies and series whenever and wherever they want without having to download the entire content beforehand. At the same time, Netflix needs to help users find something they want to watch among a very large catalog.

The platform provide another services that not will be consider in this project like: Option for subtitles, interactive experiences or content preview.

2.2 Actors

The application has 1 end user that consist in the customer that use the main function in our system. There are 3 administration users each of this person has a particular function and use the application for different purposes. Also there are some system actor like external/third-party systems that interact with application but in business processes, like provide content, money or infrastructure.

Table 1: End Users

Actor	Role	Frequency of interaction	Expectation
Customer	A person who has a monthly membership plan with the platform.	High	Be able to watch series and movies without interruptions and receive relevant recommendations.

Table 2: Administrative Users

Actor	Role	Frequency of interaction	Expectation
Operational user	A member of the operations team responsible for managing content within the system.	Medium	Be able to manage all the content without problems.
Data analyst	A member of the Data & Insights area who can access data and statistics.	Medium	Have access to useful data with clear statistics.
Platform administrator	Is responsible for managing users and the business model.	Low	Be able to manage the users of the system and membership configurations.

Table 3: External Systems and Third-Party Services

Actor	Role	Frequency of interaction	Expectation
Payment provider	An external service responsible for processing subscription payments and other financial transactions.	High	Process payments securely and provide reliable transaction confirmation.
Content provider / Production company	An external company that provides movies, series, or other audiovisual content to the platform.	Medium	Deliver licensed content and the required information reliably.
Internet Service Provider (ISP)	Provides the network connectivity through which customers access the platform and stream content.	High	Provide reliable and sufficient network connectivity for streaming.
Advertiser / Advertising partner	A company that pays to have its products, services, or brand presented to the platform's audience through advertising or branded content.	Medium	Reach the intended audience and obtain measurable exposure for its brand.

2.3 Main Use Cases

This section has the primary use cases of the application, the specification format has the identifier, description, related functional requirements, the main and alternative flows, the preconditions and post-conditions.

Table 4: Use Case CU 1.1 – Sign up in app as a customer

Use case	CU 1.1 – Sign up in app as a customer
Actors	Customer
Requirements	RF-01, RF-02
Precondition	The user does not have a registered account in the system and wishes to become a customer.
Main flow	
Actor	1. Accesses the registration screen and selects the "Sign up" option.
System	2. Displays the registration form requesting a Gmail account (or email) and password, along with the available membership plans.
Actor	3. Selects a membership plan and enters the requested account credentials.
System	4. Requests the payment method information (card or payment gateway) to process the membership.
Actor	5. Enters the payment details and confirms the payment.
System	6. Sends the payment information to the payment gateway for processing.
System	7. Validates the payment confirmation and creates the customer account.
System	8. Sends a notification confirming the account creation and active membership.
Alternative flow – Payment not accepted	
System	1. Detects that the payment gateway rejected the transaction.
System	2. Displays an error message indicating that the payment was not accepted.
Actor	3. Corrects the payment information or selects another payment method.
System	4. If no valid payment is provided, cancels the registration process and does not create the account.
Postcondition	The customer account is created with an active membership, or no account is created if the payment fails.

Table 5: Use Case CU 1.2 – Log in app

Use case	CU 1.2 – Log in app
Actors	Customer, Operational user, Data analyst, Platform administrator
Requirements	RF-03
Precondition	The user has a previously registered account with valid credentials.
Main flow	
Actor	1. Accesses the login screen.
System	2. Displays the fields to enter the Gmail account (or email) and password.
Actor	3. Enters the credentials and presses the "Log in" button.
System	4. Validates the entered credentials against the stored records.
System	5. Grants access and redirects the user to the dashboard corresponding to their role.
Alternative flow – Incorrect credentials	
System	1. Detects that the email or password entered is incorrect.
System	2. Displays an error message and increases the failed attempt counter.
Actor	3. Re-enters the credentials to try again.
System	4. If the number of failed attempts reaches 5, blocks access to the account temporarily.
Postcondition	The user is authenticated and redirected to their corresponding dashboard, or the account is temporarily blocked after 5 failed attempts.

Table 6: Use Case CU 1.3 – Update user information

Use case	CU 1.3 – Update user information
Actors	Customer, Operational user, Data analyst, Platform administrator
Requirements	RF-06, RF-04
Precondition	The user is authenticated in the system with an active session.
Main flow	
Actor	1. Accesses the "My profile" or "Account settings" section.
System	2. Displays the user's current personal information and access credentials.
Actor	3. Selects the field(s) to edit (email, password, or personal information) and enters the new values.
Actor	4. Presses the "Save changes" button.
System	5. Validates the format and consistency of the entered data.
System	6. Updates the information in the database and displays a success confirmation message.
Alternative flow – Invalid input	
System	1. Detects that one or more fields do not meet the required validation (invalid format, empty field, mismatched password, etc.).
System	2. Displays an error message specifying the invalid field(s).
Actor	3. Corrects the indicated data and resubmits the request.
Postcondition	The user's profile and/or access information is successfully updated, or remains unchanged if the validation fails.

Table 7: Use Case CU 1.4 – Cancel membership

Use case	CU 1.4 – Cancel membership
Actors	Customer
Requirements	RF-05
Precondition	The customer has an active membership and is authenticated in the system.
Main flow	
Actor	1. Accesses the "Manage membership" section and selects "Cancel membership".
System	2. Requests confirmation of the cancellation and the reason for it.
Actor	3. Confirms the cancellation and provides the requested reason.
System	4. Processes the cancellation request.
System	5. Updates the membership status to inactive and displays a confirmation notice.
Alternative flow – Login attempt after cancellation	
Actor	1. Attempts to log in after the membership has been cancelled.
System	2. Detects that the membership is inactive and denies access to the platform's content.
System	3. Notifies the customer that a new payment is required to reactivate the membership.
Postcondition	The membership is cancelled and marked as inactive; access to the platform remains restricted until the membership is renewed.

Table 8: Use Case CU 1.5 – Search content

Use case	CU 1.5 – Search content
Actors	Customer
Requirements	RF-08, RF-12
Precondition	The customer is authenticated and has an active membership.
Main flow	
Actor	1. Accesses the main screen and views the personalized recommendation feed.
System	2. Generates and displays content recommendations based on watch history, contextual data, and previous searches.
Actor	3. Enters a search query in the search bar or selects a title from the recommended feed.
System	4. Searches the content catalog for titles matching the query.
System	5. Displays the list of matching movies/series along with their metadata (title, genre, synopsis, image).
Alternative flow – No exact match found	
System	1. Detects that there are no exact matches for the entered search query.
System	2. Generates and displays a list of similar or related content based on the recommendation engine.
Actor	3. Selects one of the suggested titles to continue.
Postcondition	The customer receives a list of content matching or related to their search, ready to be selected for playback.

Table 9: Use Case CU 1.6 – Plays content

Use case	CU 1.6 – Plays content
Actors	Customer
Requirements	RF-07
Precondition	The customer is authenticated, has an active membership, and has selected a movie or episode from the catalog.
Main flow	
Actor	1. Selects the movie or episode to watch and presses "Play".
System	2. Loads the selected content and starts instant video streaming without requiring a prior download.
System	3. Displays the playback controls (play, pause, rewind, forward, volume).
Actor	4. Watches the content, using the playback controls as needed.
Alternative flow 1 – Video streaming failure	
System	1. Detects a failure in the video stream (e.g., corrupted file or server error).
System	2. Pauses the playback and displays an error message explaining the issue.
Actor	3. Chooses to retry playback or select different content.
Alternative flow 2 – Unstable internet connection	
System	1. Detects that the user's internet connection is slow or unstable.
System	2. Displays a buffering/loading indicator while attempting to recover the stream.
Actor	3. Waits for the stream to resume or checks their internet connection.
Postcondition	The content is streamed successfully and watched by the customer, or playback is interrupted and the customer is properly notified of the issue.

Table 10: Use Case CU 1.7 – Manage content

Use case	CU 1.7 – Manage content
Actors	Operational user
Requirements	RF-09
Precondition	The operational user is authenticated and has permissions to manage the content catalog.
Main flow – Upload new content	
Actor	1. Accesses the content management module and selects "Add new content".
System	2. Displays a form to upload the video file and enter its metadata (title, genre, description, cast, duration).
Actor	3. Uploads the video file and completes the requested metadata.
System	4. Validates the uploaded information and stores the new content in the catalog.
System	5. Displays a confirmation message and updates the catalog state.
Alternative flow 1 – Edit existing content	
Actor	1. Selects a specific video already registered in the catalog.
Actor	2. Modifies the video file and/or its metadata.
System	3. Validates and updates the record, then confirms the change and refreshes the catalog state.
Alternative flow 2 – Delete content	
Actor	1. Selects a specific video and requests its deletion.
System	2. Performs a logical deletion, marking the video as inactive without losing its metadata.
System	3. Confirms the deletion and updates the catalog view.
Postcondition	The content catalog is updated with the new, modified, or logically deleted item, without permanent loss of metadata.

Table 11: Use Case CU 1.8 – Manage users

Use case	CU 1.8 – Manage users
Actors	Platform administrator
Requirements	RF-10
Precondition	The platform administrator is authenticated and has permissions to manage user accounts and roles.
Main flow – Register a new internal user	
Actor	1. Accesses the user management module and selects "Add user".
System	2. Displays a form to register the user's data and assign a role (operational user, data analyst, or administrator).
Actor	3. Enters the user's information and selects the corresponding role.
System	4. Creates the account, assigns the selected role, and confirms the operation.
Alternative flow 1 – Change a user's role	
Actor	1. Selects an existing user from the list.
Actor	2. Changes the role currently assigned to the user.
System	3. Updates the user's role and permissions, then confirms the change.
Alternative flow 2 – Ban a user	
Actor	1. Selects a user and requests to ban the account.
System	2. Requests a reason for the ban.
Actor	3. Enters the reason for banning the user.
System	4. Disables the user's account, records the reason, and confirms the action.
Postcondition	The user account is created, its role is updated, or the account is banned according to the administrator's action.

Table 12: Use Case CU 1.9 – View reports

Use case	CU 1.9 – View reports
Actors	Data analyst
Requirements	RF-11
Precondition	The data analyst is authenticated and has access to the analytics module.
Main flow	
Actor	1. Accesses the reports/analytics module.
System	2. Displays the available report types and filter options (time range, metric).
Actor	3. Selects the desired report and configures the filter parameters.
System	4. Processes the streaming usage data according to the selected filters.
System	5. Displays the processed analytical report with the corresponding metrics.
Alternative flow – Report cannot be displayed	
System	1. Detects an error while processing or generating the requested report.
System	2. Notifies the data analyst that the report could not be displayed.
Actor	3. Retries the request or selects a different report/filter configuration.
Postcondition	The requested analytical report is displayed to the data analyst, or the analyst is notified that it could not be generated.

2.4 System Boundaries

Subsection example 2

3 Functional Requirements

This requirements specify what application does. There are 12 functional requirements divided in 7 domains. To define the functional scope for this section, an analysis of Netflix’s business model and official domain vision was conducted to establish a comprehensive understanding of the operational context. Building upon this business domain baseline, an actor-driven requirement elicitation process was executed. System actors, ranging from end-customers to operational staff and data analysts, were identified and evaluated based on their operational responsibilities and system interactions. Each actor’s functional needs, such as secure authentication, uninterrupted media streaming, content curation, and platform analytics, were systematically mapped to define the required system capabilities.

Subsequently, Domain-Driven Design (DDD) strategic design principles were applied to analyze and categorize the elicited requirements. The requirements were grouped into distinct Bounded Contexts, acknowledging that domain entities and ubiquitous language terms carry specialized business semantics

and rules depending on their contextual boundary. Through this domain decomposition, the following core Bounded Contexts were identified:

- **User Account Management:** Manages user accounts, authentication credentials, and access control policies across the platform.
- **Subscription Management:** Governs membership plans, billing processing, and subscription lifecycle states.
- **Streaming Core:** Serves as the primary domain service responsible for video playback orchestration and real-time media streaming delivery.
- **Recommendation Engine:** Leverages user consumption history, search queries, and contextual metadata to generate personalized content suggestions.
- **Content Management:** Facilitates the complete lifecycle management (CRUD operations) for media assets and metadata across staging and release workflows.
- **Data & Insights:** Captures, aggregates, and processes platform-wide performance and streaming metrics for analytical insights.
- **Catalog & Discovery:** Handles content indexing, search queries, and catalog exploration to enable seamless user discovery.

Table 13: System Requirements Matrix Categorized by Domain / Bounded Context

ID	Requirement (The system shall...)	Inputs	Outputs	Domain / Context
RF - 01	Allow customer registration through paid membership.	Email, password, payment details	Customer account creation notification	Subscription Management (Subscription)
RF - 02	Allow membership payment processing using online payment methods.	Payment method info (card/gateway)	Payment confirmation or error message	
RF - 05	Allow customers to cancel their active membership subscription.	Cancellation request and reason	Membership cancellation status confirmation	
RF - 03	Allow any user to authenticate using email and password credentials.	Email and password credentials	Access granted to appropriate user dashboard	User Account Management

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ID	Requirement (The system shall...)	Inputs	Outputs	Domain / Context
RF - 04	Allow users to update their access credentials.	Registered email, current and new credentials	Credential update success or failure notice	(User Account Management)
RF - 06	Allow any user to update their personal profile information.	Authenticated session, updated attributes	Updated profile status or error message	(User Account Management)
RF - 10	Allow platform administrators to manage user accounts and roles.	Admin request parameters and action payload	Success/failure message and updated user state	(User Account Management)
RF - 07	Allow customers to stream movie or episode content instantly without prior downloading.	Chapter/movie selection	Instant video playback stream	Streaming Core
RF - 08	Provide personalized and timely content recommendations to customers.	Contextual data, watch history, search queries	Personalized content feed on main screen	Recommendation Engine
RF - 09	Allow operations personnel to manage media content catalog items.	Metadata creation, update, query, or deletion requests	Action result message and updated catalog state	Content Management
RF - 11	Allow data analysts view streaming usage and others reports.	Selection of the report	Processed analytical reports and metrics	Data & Insights
RF - 12	Allow customers to search for movies or series in the catalog.	Search query string	List of matching movies/series and metadata	Catalog & Discovery

4 Non-Functional Requirements

The non-functional requirements (NFRs) defined for the system are based primarily on publicly available information regarding the Netflix platform and its operational characteristics. The purpose of these requirements is to establish a realistic reference workload and performance context for the cloud-native prototype developed in this project. Some of the values presented correspond to reported Netflix figures, while others are estimates derived from those publicly available values using explicit calculations and assumptions.

In cases where an official value was not directly available, an estimate was derived from the available

information rather than introducing an arbitrary value. This approach allows the prototype to preserve a relationship with the real operational scale of Netflix while maintaining transparency regarding which values are directly documented and which are calculated approximations.

It is important to emphasize that the functional prototype developed for this project will not reproduce the full production scale of Netflix.

If the prototype remains stable under the simulated workload and demonstrates the expected behavior in terms of concurrency, request processing, availability, data management, and scalability, the architecture can subsequently be scaled horizontally by increasing the computational and infrastructure resources provided by the cloud platform.

The objective is not to claim that the prototype can directly sustain Netflix-scale traffic with academic-level infrastructure, but to demonstrate that the system can maintain stability under representative workloads and that its architecture provides a practical foundation for increasing capacity through horizontal scaling as additional cloud resources become available.

Table 14: RNF – Expected Users

Identifier	Description	RF associated
RNF-01	The system shall support at least 325,000,000 customer memberships while maintaining normal operation and the defined performance targets. This value corresponds to the reported number of paid memberships reached during 2025. <i>Reference:</i> Source	RF-01, RF-03
RNF-02	Assuming the 96 billion hours viewed in the second half of 2025 were uniformly distributed across its 184 days: $\frac{96,000,000,000 \text{ h}}{184 \times 24 \text{ h}} \approx 21.74 \text{ million concurrent users}$ Therefore, the system shall support at least 22 million concurrent users under normal conditions. <i>Reference:</i> Source	RF-03, RF-07, RF-08, RF-12
RNF-03	The system shall support at least 65 million concurrent users during peak activity. This value is an estimate based on the highest publicly reported concurrent-stream peak and shall not be interpreted as an officially stated maximum system capacity. <i>Reference:</i> Source	RF-03, RF-07, RF-08, RF-12

Table 15: RNF – Average Requests per Hour

Identifier	Description	RF associated
RNF-04	The system shall support approximately 677,000,000 application requests per hour during normal operating conditions. This is an estimate based on a historical public figure of more than 2 billion incoming API calls per day for more than 40 million members, corresponding to more than 50 API calls per member per day. Applying this ratio to 325 million memberships gives approximately 16.25 billion calls per day, or approximately 677 million calls per hour. <i>Reference:</i> Source	RF-03, RF-06, RF-08, RF-09, RF-10, RF-12

Table 16: RNF – Peak Requests per Hour

Identifier	Description	RF associated
RNF-05	The system shall support approximately 2.02 billion application requests per hour during peak periods. This is an estimate obtained by scaling the normal request rate according to the ratio between peak and normal concurrent users: $677.08 \text{ M requests/h} \times \frac{65 \text{ M concurrent users}}{21.74 \text{ M concurrent users}} \approx 2.02 \text{ B requests/h}$ This calculation assumes that application request volume scales approximately with concurrent activity. <i>Reference:</i> Peak reference	RF-03, RF-07, RF-08, RF-12

Table 17: RNF – Peak Duration

Identifier	Description	RF associated
RNF-06	The system shall sustain the defined peak workload continuously for at least 5 hours. This requirement is derived from the reported peak-duration range of approximately 3 to 5 hours, with the upper bound used as the workload target.	RF-03, RF-07, RF-08, RF-12

Table 18: RNF – Response-Time Targets

Identifier	Description	RF associated
RNF-07	At least 95% of authentication, profile, subscription, and standard API requests shall receive a response within 500 ms under normal operating conditions. This is an estimated end-to-end application target informed by public high-scale service measurements, where individual backend services operate with substantially lower internal latencies. <i>References:</i> Source 1; Source 2	RF-01, RF-02, RF-03, RF-04, RF-05, RF-06
RNF-08	At least 95% of authentication, profile, subscription, and standard API requests shall receive a response within 1,000 ms during peak operating conditions. This peak target is estimated as twice the normal response-time target to provide bounded latency under increased queueing and contention.	RF-01, RF-02, RF-03, RF-04, RF-05, RF-06
RNF-09	At least 95% of catalog browsing and content-search requests shall return the initial result set within 1,000 ms under normal operating conditions. This is an estimated target based on high-scale systems that use low-latency data retrieval and precomputed or cached results for interactive user experiences. <i>Reference:</i> Source	RF-09, RF-12
RNF-10	At least 95% of catalog browsing and content-search requests shall return the initial result set within 2,000 ms during peak operating conditions. This peak target is estimated as twice the normal response-time target.	RF-09, RF-12
RNF-11	At least 95% of recommendation requests shall return a personalized content feed within 2,000 ms under normal operating conditions. This is an estimated application-level target; public engineering documentation confirms that online recommendation computation is subject to explicit response-time SLAs and therefore requires bounded real-time execution and fallback mechanisms. <i>Reference:</i> Source	RF-08
RNF-12	At least 95% of recommendation requests shall return a personalized content feed within 4,000 ms during peak operating conditions. This peak target is estimated as twice the normal response-time target while preserving bounded response latency under increased demand.	RF-08
RNF-13	At least 95% of successful video playback requests shall display the first video frame within 2,500 ms under normal operating conditions, excluding delays caused by the user’s local network or Internet service provider. This target is based on a published video-platform KPI that defines time-to-first-video-frame below 2,500 ms. <i>Reference:</i> Source	RF-07
RNF-14	At least 95% of successful video playback requests shall display	RF-07

Table 19: RNF – Availability

Identifier	Description	RF associated
RNF-15	The system shall maintain a monthly availability of at least 99.99% for customer-facing services. This corresponds to the four-nines availability target publicly reported for large-scale operations. <i>Reference:</i> Source	RF-03, RF-07, RF-08, RF-09, RF-12
RNF-16	The maximum accumulated unplanned downtime of customer-facing services shall not exceed approximately 4.3 minutes per 30-day month. This value is derived directly from the 99.99% monthly availability target: $30 \times 24 \times 60 \times 0.0001 = 4.32 \text{ minutes} \approx 4.3 \text{ minutes}$	RF-03, RF-07, RF-08, RF-09, RF-12

Table 20: RNF – Data Volume

Identifier	Description	RF associated
RNF-17	The system shall support at least an estimated 420 TB of application-managed data at launch, including user accounts, subscriptions, content metadata, viewing history, recommendation data, search records, and analytics data. This estimate is derived from a comparable large-scale streaming platform that reports petabytes of data residing in Bigtable for online use cases and 777 million monthly active users. Taking 1 PB as a conservative lower bound: $1 \text{ PB} \times \frac{325 \text{ M users}}{777 \text{ M users}} \approx 418 \text{ TB} \approx 420 \text{ TB}$ This is a workload estimate, not a published storage capacity. <i>Reference:</i> Source	RF-01, RF-02, RF-05, RF-06, RF-08, RF-09, RF-11, RF-12
RNF-18	The system shall support a media catalog of at least 8,500 movie and TV-series titles, including the metadata required for catalog discovery and playback. This value is a conservative approximation based on a publicly indexed national catalog containing approximately 8,505 titles. <i>Reference:</i> Source	RF-07, RF-09, RF-12

Table 21: RNF – Data Growth

Identifier	Description	RF associated
RNF-19	The system shall be designed to accommodate an estimated gross data-generation rate of up to 104,000 TB per year through scalable storage and data-lifecycle management. The estimate is derived from an observed generation rate of approximately 2 PB per week: $2 \text{ PB/week} \times 52 \text{ weeks/year} \approx 104 \text{ PB/year} = 104,000 \text{ TB/year}$ This value represents gross data generated in production workflows rather than net persistent storage, since a significant portion of generated data may subsequently be deleted or become unused. <i>Reference:</i> Source	RF-07, RF-08, RF-09, RF-11, RF-12

Table 22: RNF – Geographic Scope

Identifier	Description	RF associated
RNF-20	The system shall initially provide its customer-facing services to users located in Colombia through the public Internet.	RF-01, RF-02, RF-03, RF-07, RF-08, RF-09, RF-12
RNF-21	The system architecture shall permit expansion from the initial Colombian deployment to more than 190 country and regional contexts without requiring changes to the core application model. As a reference for the required geographic scale, a published large-scale streaming infrastructure footprint comprises 18,000 servers distributed across 6,000 locations in 175 countries. <i>References:</i> Source 1; Source 2	RF-01, RF-02, RF-03, RF-07, RF-08, RF-09, RF-12

5 Workload Model

Structured table with all workload parameters and justification for each value

References

- [1] Rummary, G. A., & Niranjana, M. (1994). On-line Q-learning using connectionist systems. *Technical Report*, Cambridge University Engineering Department.